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23 May 1978
SUPERSEDING
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30 July 1973

MILITARY SPECIFICATION

TOPPING, DESSERT AND BAKERY PRODUCTS, FROZEN OR DEHYDRATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers frozen topping and dehydrated topping (requiring reconstitution) for desserts and bakery products for use by the Armed Forces as an item of general issue (see 6.3).

1.2 Classification.

1.2.1 Types and classes. Toppings covered by this specification shall be of the following types and classes as specified (see 6.1).

Type I - Frozen

Class 1 - With nonfat milk solids
Class 2 - Without nonfat milk solids

Type II - Dehydrated

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

C-M-350	-	Milk, Nonfat, Dry.
TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, ATTN: DRDNA-ES, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8940

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PPP-B-1163 - Box, Corrugated Fiberboard, High Compression, Strength, Weather-Resistant, Wax-Resin Impregnated.
* PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.
* PPP-C-96 - Cans, Metal 28 Gage and Lighter.

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MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-L-35078 - Loads Unit: Preparation of Nonperishable Subsistence.

STANDARDS

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

US DEPARTMENT OF AGRICULTURE

General Specifications for Dairy Plants Approved for USDA Grading and Inspection Service
* US Standards for Dry Whey

(Application for copies should be addressed to the Dairy Section, Standardization Branch, Poultry and Dairy Quality Division, Room 2945-S, Food Safety and Quality Service, US Department of Agriculture, Washington, DC 20250.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

INTERNATIONAL ASSOCIATION OF MILK AND FOOD SANITARIANS, INC.

3-A Sanitary Standards

(Application for copies should be obtained from the International Association of Milk and Food Sanitarians, Inc., Box 701, Ames, IA 50010.)

AMERICAN PUBLIC HEALTH ASSOCIATION

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1015 18th Street, N.W., Washington, DC 20036.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN OIL CHEMISTS' SOCIETY

Official and Tentative Methods of the American Oil Chamists' Society.

(Application for copies should be addressed to the American Oil Chemists' Society, 508 South Sixth Street, Champaign, IL 60601.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of this product shall conform to the requirements of table I.

* TABLE I. Material requirements

Item	Requirement
1	Fat shall be refined deodorized vegetable fat or oil and may be hydrogenated in varying degrees. All fat shall be handled under strictly sanitary conditions and shall be bland, sweet, clean, and free from oxidized, rancid, tallowy, or other objectionable tastes and odors. Component fats or oils may contain antioxidants and emulsifiers provided the antioxidants and emulsifiers are in accordance with items 2 and 3, respectively. The vegetable fat or oil shall have a free fatty acid content, expressed as oleic acid, not in excess of 0.05 percent and a maximum peroxide value of 0.5 meq of peroxide per kilogram of product.
2	Antioxidants incorporated into the fat or oil shall be of the kind(s) and at levels permitted in accordance with regulations of the Food and Drug Administration promulgated under the Food Additive Amendment concerning rendered animal fats, processed vegetable fat or oil, or blends thereof.
3	Emulsifying agents shall be commercial monoglyceride concentrates or other suitable emulsifying agents which are permitted by the Food and Drug Administration for use in edible fats or oils.
4	Nonfat milk solids shall be derived from either nonfat dry milk, type I, style B or C of C-M-350 or from skim milk meeting items 1 and 2 of table I, section 3.1 of C-M-350. The nonfat dry milk shall be not more than 60 days old at the time of manufacture of the end item.
* 5	Protein, food grade, (for use in type I, class 2 and type II products only) shall be a product derived from plant sources or by special treatment of milk solids to produce a solubilized caseinate preparation. In addition, sweet whey solids shall be US Extra Grade at the time of use and shall not exceed 90 days after the process of spray drying.

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TABLE I. Material requirements (cont'd)

Item	Requirement
6	Sweetening ingredients, either in dry or in liquid form shall be of food grade quality and shall be sucrose, lactose, corn sugar, corn syrup solids, invert sugar, or mixtures thereof.
7	Active flavoring ingredients shall be pure vanilla extract, imitation vanilla, or a combination of both.
8	Salt, if used, shall be white refined, free flowing sodium chloride, with or without anti-caking agent.
9	Coloring material shall be vegetable coloring or U.S. certified colors.
10	Anticaking agents, if used, to enhance the free flowing properties (restricted to type II product only) shall be of the type and amounts used which meet the requirements of the Federal Food, Drug, and Cosmetic Act.

3.2 Processing.

3.2.1 Type I, class 1 product. As applicable, all incoming fresh milk or skimmed, unless processed within 2 hours, shall be cooled immediately to 45°F (7°C) or less.

3.2.2 Type I, classes 1 and 2. Using good commercial practices, all applicable components, including sweeteners, shall be combined and, except for flavoring (to be added after heat processing and cooling), the entire mix shall be pasteurized by heating to a temperature of 155°F (68°C) and holding at this temperature for 30 minutes, or by other recognized methods of pasteurization acceptable to local and state health authorities which give equivalent results. All pasteurization equipment, sanitary design, and construction standards shall conform to applicable 3-A Sanitary Standards and Accepted Practices. The product shall be homogenized and it shall be frozen immediately after packaging to a temperature of below 0°F (-18°C) and held at this temperature until shipped.

3.2.3 Type II product. The entire mix, when combined as in 3.2.2, shall be spray dried in accordance with good industry practice (a maximum outlet temperature of 185°F (85°C) is recommended to insure compliance with 3.3) or the finished product may be manufactured by blending of the dry ingredients.

3.3 Finished product requirements. The finished product shall meet the requirements of table II.

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TABLE II. Finished product requirements

Item	Requirement	Type I	Type II
	Physical requirements All types 1/		
1	The color shall be uniformly white to light cream color.		
2	The product shall be uniform in composition and free from foreign material such as glass, paint, wood, metal, rust, hair, rodent evidence, insects, insect parts or other extraneous material.		
3	The product shall be characteristically sweet, clean, with a clean pleasant vanilla taste and odor and shall be free from oxidized, stale, cooked, rancid or other objectionable tastes and odors.		
	Type II product		
4	The product shall be free from lumps that do not fall apart readily with light finger pressure.		
	Analytical requirements	Type I	Type II
5	Total fat not less than (percent)	20	34
6	Moisture, not more than (percent)	50	1.50
7	Phosphatase activity 2/	Less than 1 mcg of phenol per ml of product	Less than 1 mcg of phenol per ml of reconstituted product
8	Scorched partial content, mg, not more than	-	1.50
	Microbiological requirements		
9	Standard plate count, not more than 10,000 per g		
10	Coliform count, not more than 10 per g		
11	Salmonellae test	Negative	

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TABLE II. Finished product requirement (cont'd)

Item	Requirement
	Performance requirements (see 4.5.4)
12	The product shall whip to an overrun of 175 percent (minus tolerance of 15 percent) within 10 minutes after whipping action (high speed) is started. (NOTE: 10 minutes is maximum time permitted; overrun may be reached in less time).
13	The whipped product shall show no serum separation or weeping in 2 hours at 70°F to 80°F (21°C to 26°C).
	Whipped product physical requirements
14	The whipped product shall have a white to light cream color, a stiff body, dry appearance and a smooth creamy texture. The whipped product shall not be frothy or fluffy or break down in the emulsion when whipped within the 10 minute limitation and shall be characteristically sweet with a vanilla flavor background and free from objectionable tastes and odors (see item 3, above).

1/ For type I, examine after defrosting at 40°F to 45°F (4°C to 7°C).

2/ The phosphatase test (by the Scharer Rapid Method) (see 4.5.3) shall be performed only when fresh dairy products (e.g, skim milk) are used in formulation.

3.4 Age and temperature of product (type I only) at time of delivery. The product shall be delivered within 120 days after date of manufacture. The temperature of type I product, at time of delivery; shall not exceed 10°F (-12°C).

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Plant qualification. The product shall originate, be processed, marked and handled in plants which are operated under the requirements of MIL-STD-668, or in dairy plants approved for USDA Inspection and Grading Service and listed by the US Department of Agriculture in the document titled: Dairy Plants Surveyed and Approved for USDA Grading Service.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.6.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients. Conformance to the applicable identity and condition requirements specified in table I shall be determined in accordance with the examination criteria in 4.3.1.1.1 and 4.3.1.1.2. One or more nonconformances shall be cause for rejection of the component or the finished product made therefrom, as applicable.

4.3.1.1.1 Identity. Conformance to the applicable identity requirements, kinds, methods of manufacture, additives, etc., shall be ascertained by examination of labels, invoices or other valid documents.

4.3.1.1.2 Condition. For determining conformance to condition requirements as may be contained in the applicable ingredient requirement of table I, the sample for organoleptic examination, as necessary, shall be approximately 1 pound (454 g) of product derived from five containers or all containers if five or less form a lot.

* 4.3.1.1.3 Foil laminated bag. Conformance of bag material to requirements in 5.1.3 shall be ascertained by examination of invoices or other valid documentation from the contractor of the envelope material. Nonconformance shall indicate unacceptable material and use of such shall be cause for rejection of the end item.

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4.3.1.2 Testing of components. Testing of components described in this specification shall be as shown in table III. Result for tinplate shall be applicable to the lot average and pass or fail for free fatty acids.

TABLE III. Testing of components (see 4.3.1)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref & rqmt para
Cans and lids 1/	One can and lid	Cans	S-1	Tinplate	Nearest 0.01 lb/ base box	4.5.5 5.1.1
Vegetable fat or oil	One pint (0.48 liters) composite	Container	S-1	Free fatty acid	Oleic acid 0.01 percent peroxide value 0.1/ meq/kg	3.1 Item 2 4.5.1

1/ Exterior can coating, when required, and can labeling shall be tested in accordance with the applicable subsidiary specifications except that the tests shall be performed on the same cans submitted for tinplate testing.

4.3.2 In-process examination. Examination shall be made to determine compliance with the requirements of 3.2. Time and temperature records shall be maintained. Nonconformance to the above referenced requirements reflected by actual examination or by records shall be cause for rejection of the component lot or the finished product made therefrom, as applicable.

4.3.3 Examination of end item. Classification of defects found during examinations shall be in accordance with tables V through VII and 4.3.3.1 through 4.3.3.5. The examination criteria for the above referenced tables shall be in accordance with table IV. The acceptable quality levels (AQLs) shall be expressed as percent defective for table VI and as defects per hundred units for tables V and VII.

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TABLE IV. End item examination criteria

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major A	Major B	Minor
V	One primary container	Primary containers	S-3	0.40	1.5	4.0
VI	One primary container	Primary containers	S-3	--	--	2.5
VII	One primary container	Primary containers	S-3 (applicable to type II product) S-2 (applicable to type I product)	1/	1.5	--

1/ The finding of any Major A defect shall be cause for rejection of the lot.

TABLE V. External examination of cans

Category			Defect
Major A	Major B	Minor	
101			Leaker
102			Springer, flipper or sweller
103			False seam or improperly closed
	151		Not type, style or size specified
	152		Severe dent
	153		Dent causing sharp ridge
	154		Severe paneling, buckling or collapsed can
	155		Cable cut
	156		Pitted rust 1/
		201	Not clean 2/
		202	Moderate dent
	157		Bulging ends more than 1/4-inch (6 mm) beyond lip 3/

1/ Rust that can be removed by wiping with a soft cloth will not be considered as pitted rust.

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2/ Cans showing a very thin film of grease which is discernable to the touch but not readily discernable visually are considered to be clean.

3/ Applicable to the frozen product.

TABLE VI. Examination of net weight 1/

Category	Defect
Minor	
201	Net weight less than 15-3/4 ounces (446 g) 2/
202	Net weight less than 6 lbs. 7-1/2 ozs (2.93 kg) or 6 lbs 15-1/2 ozs (3.16 kg) (as applicable) 3/

1/ The lot shall be rejected if the average net weight of the samples is less than the net weight specified.

2/ Weigh to a tolerance of 1/8-ounce (3.5 g).

3/ Weigh to a tolerance of 1-ounce (28 g).

TABLE VII. Examination of product characteristics

Category		Defect
Major A	Major B	All types
101		Presence of foreign material, e.g., glass, paint, wood, metal, rust, hair, insects or insect fragments, rodent evidence or other extraneous matter.
	151	Not a sweet, clean, characteristic vanilla taste or odor.
102		Presence of oxidized, stale, cooked, rancid or other objectionable tastes or odors.
		Applicable to type II product
	152	Presence of lumps that do not fall apart readily with light finger pressure.

4.3.3.1 Examination of can coating. Examination of can coating, when applicable, shall be in accordance with the examination criteria of TT-C-495.

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4.3.3.2 Examination of labeling. Examination of labeling shall be in accordance with the examination criteria of MIL-L-1497.

4.3.3.3 Examination of unit loads. Examination of unit loads shall be in accordance with the examination criteria of MIL-L-35078.

* 4.3.3.4 Examination of the filled and closed shipping container for level A, B or C. Examination of the filled and closed shipping container shall be in accordance with PPP-C-29. In addition, the following defects shall be included in the table of examination: Major; marking, missing, incorrect or illegible. When alternative level B pack is selected, the examination of the filled and closed shipping container shall be in accordance with the examination criteria of PPP-B-1163.

4.3.3.5 Examination of waterproof case liner. When a waterproof case liner is specified, the examination criteria shall be in accordance with MIL-L-10547.

4.3.3.6 Examination for leakage. Examination shall be in accordance with the examination criteria of PPP-C-29.

4.3.4 Sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested for fat, moisture, phosphatase activity (see footnote 2/ to table II), microbiological requirements for Standard Plate Count, coliform count and Salmonella test and the performance requirements of item 11 through 13 of table II. Procedures for testing shall be in accordance with 4.5.1, 4.5.2, 4.5.3, 4.5.4.1 and 4.5.4.2. Test requirements shall be lot average requirements except for microbiological requirements which shall be unit testing. The sample for microbiological testing shall be one filled and closed can. The sample size shall be the number of cans indicated by inspection level S-2. Lot size shall be expressed in terms of the sample unit. The sample for all other test requirements shall be an approximate 1-pound (454 g) composite derived from the number of sample units required for microbiological testing. The acceptable quality levels (AQL's)s) expressed as percent defective for each microbiological characteristic, except Salmonella, shall be 1.5. Results of the microbiological counts shall be reported in accordance with Standard Methods for the Examination of Dairy Products; Salmonella shall be reported as pass or fail; analytical results shall be reported to the same unit or decimal as specified for the pertinent requirements; whipping test (overrun) shall be reported to the nearest one percent overrun and serum separation as pass or fail. One or more test failures, as applicable, shall be cause for rejection of the lot.

4.4 Examination of age and temperature of product (type I) at time of delivery. Examination shall be made to determine compliance with the requirements of 3.4 as concerns age and temperature at time of delivery.

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Ascertainment of age shall be determined by examination of markings of the filled and closed shipping container. The examination criteria for determining conformance with the temperature requirements shall be as follows: lot size shall be expressed as shipping containers, the sample unit shall be one filled and closed shipping container. The inspection level shall be S-3. The AQL expressed as defects per hundred units shall be 4.0. In addition, nonconformance to the age of product shall be cause for rejection of the lot.

4.5 Test methods.

4.5.1 Chemical analyses. The following chemical analyses shall be made in accordance with the referenced methods:

Test	Section	Methods
Component:		
Vegetable fat or oil Free fatty acids	(Commercial Fats and Oils)	Ca 5a - 40 1/
Peroxide value	Ibid.	Cd 8 - 53 1/
Finished product:		
Type I Fat	Ice Cream and Frozen Desserts	Fat 2/
Moisture	Ibid.	Total solids 2/
Type II Fat	Dried Milk, Nonfat Milk and Malted Milk	Fat in dried milk 2/
Moisture	Ibid.	Moisture 2/

1/ Official and Tentative Methods of the American Oil Chemists' Society.

2/ Official Methods of Analysis of the Association of Official Analytical Chemists. Type I samples shall be thoroughly defrosted and tempered to 40°F to 45°F (4°C to 7°C) prior to sampling and testing. The total solids test shall be subtracted from 100.00 percent to ascertain moisture value.

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4.5.2 Microbiological analyses. Microbiological analyses shall be made in accordance with the following methods in Standard Methods for the Examination of Dairy Products unless otherwise specified:

Item	Test	Method
1	Standard plate count	Agar Plate Method
2	Coliform count	Coliform Bacteria, using Violet Red Bile Agar
3	Salmonellae	1/

1/ Salmonellae test procedures shall be conducted as outlined in the Official Methods of Analysis of the Association of Official Analytical Chemists: Chapter; Microbiological Methods; Section, Salmonella. Proceed as for dried egg, except using nutrient broth instead of lactose broth in initial dilution.

4.5.3 Phosphatase test (as applicable, see item 7, footnote 2/ to table II). This test shall be performed using the Scharer Rapid Method as described in the chapter, "Phosphatase Methods to Determine Pasteurization" of Standard Methods for the Examination of Dairy Products. Follow method as used for cream.

4.5.4 Performance test. Performance of the product shall be tested by the following methods:

* 4.5.4.1 Whipping test for overrun determination.

(a) Ambient air temperature shall be 60°F to 80°F (16°C to 27°C).

(b) The whipping test shall be performed using a Hobart Model 4C or equivalent mixer, equipped with the standard 3-quart (2.8 liters) stationary pyrex or stainless steel bowl and with an all purpose beater. 1/

(c) A clean, dry, chilled 40°F to 45°F (4°C to 7°C) whipping bowl shall be used.

(d) A 1 pint (0.47 liter) measuring container shall be used for volume and weight determination. Tare weight will be deducted when performing calculations in (i) below. All measurements of unwhipped and whipped product shall be based on flush filling of the pint containers making certain that any product spilled on the sides of the container is wiped off. When filling the container with whipped product special care must be taken to: (1) eliminate air voids and, (2) avoid expelling incorporated air by pressing down with spoon or spatula using undue pressure, and (3) level off with a spatula to a flush fill.

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(e) Type I product sample preparation shall require the complete and uniform thawing and tempering of the product to between 40°F and 45°F (4°C to 7°C).

(f) Type II product sample preparation shall be in accordance with the manufacturer's instructions except that for purposes of this test, blending of the mix and water will be by hand using a large spoon or rubber spatula to avoid incorporation air. All components shall be uniformly mixed so that there is no visible evidence of dry powder or lumps in the whipping bowl. Reconstituted product will be chilled to between 40°F and 45°F (4°C to 7°C).

(g) Place 1 pint (0.47 liters) of product in the chilled whipping bowl. Start the whipping at dial setting No. 2 (low speed) to avoid splashing. Shift to No. 6 (medium speed) gradually. When a noticeable volume increase has been obtained, finish whipping at speed No. 10 (high speed) until standing peaks are formed. Stop whipping action and record high speed whipping time. 2/

(h) If high speed whipping time is less than 10 minutes, perform overrun calculation (see (i) below). If minimum overrun of 175 percent has not been reached, return sample to whipping bowl and resume high speed whipping for the balance of the 10 minutes remaining and then perform the overrun calculation. Two whipping tests will be performed from different samples and the results averaged for determining conformance to overrun requirement. If individual results show a difference of more than 25 percent overrun the procedure shall be repeated.

(i) Calculate the overrun as follows:

$$\text{Percent overrun} = \frac{A - B}{B} \times 100$$

Where A = Weight of unit volume (1 pint) (0.47 liters) of topping at 40°F to 45°F (4°C to 7°C) prior to whipping.

B = Net weight of the same volume of whipped topping.

1/ If mixing bowl is too small to accommodiate the full amount of powder and reconstituted nonfat milk, use a smaller but proportionate amount of both components so that the fully whipped product does not exceed approximately three quarters of the bowl capacity.

2/ If the dial settings are not designated numerically, the substitute of low, medium, and high speed controls will be deemed acceptable.

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4.5.4.2 Serum drainage test. Whip product using procedure in 4.5.4.1, but whip to a point where the standing peaks are stiff and relatively dry. Fill an 8-ounce (240 ml) standard kitchen measuring cup to capacity avoiding air pockets or voids. Level off with spatula to a flush fill point and transfer contents of the cup in a single heap onto a 6-inch (152 mm) square of No. 26 B & S gauge 16 mesh woven wire, placed over an appropriate size funnel and supported by a funnel holder on a ring stand. The whipped product after 2 hours exposure to a room temperature of 70°F to 80°F (21°C to 27°C) shall show no weeping or serum separation as evidenced by an absence of liquid which if present would be collected in a graduated cylinder placed under the funnel.

* 4.5.5 Determination of scorched particles. The determination of scorched particles in the product shall be performed with the procedure described in Methods of Laboratory Analysis for Dry Whole Milk and Nonfat Dry Milk, DA Instruction No. 918-103-1.

4.5.6 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation-packaging. The product shall be packaged in accordance with level A, B or C as specified (see 6.1).

5.1.1 Level A. One pound (454 g) of the type II product shall be packaged into a size 401 by 411, 401 by 508 or 509, or 404 by 404, open top style, round metal can with soldered side seam and compound-lined, double-seamed ends. A minus 1/4-ounce (7 g) tolerance will be allowed in any one container provided the average net weight of the cans is not less than one pound (454 g). The can shall be made throughout from not less than commercial 0.25 pound (113 g) electrolytic tin plate per base box (see 4.5.6). The can shall be coated overall outside with a coating conforming to type I of TT-C-495. The can shall be hermetically sealed and shall not leak when examined in accordance with 4.3.3.6.

5.1.2 Level B. Six and one half (2.95 kg) or seven pounds (3.2 kg) of the type I product shall be filled into a size 603 by 700 or 708 open-top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. A minus 1/2-ounce (14 g) tolerance will be allowed in any one container provided the average net weight of the cans is not less than 6-1/2 (2.95 kg) or 7 pounds (3.18 kg) as applicable. The can shall be made throughout from not less than commercial 0.25 pound (113 g) electrolytic tinplate per base box (see 4.5.6). The can shall be coated overall outside with a coating conforming to type I or III of TT-C-495. The can shall be hermetically sealed and shall not leak when examined in accordance with 4.3.3.6. The leakage examination shall be performed subsequent to the packaging operation and prior to the freezing operation.

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5.1.3 Level C. The type I and II product shall be packaged in accordance with 5.1.1 or 5.1.2, except that cans with or without commercial exterior coating will be acceptable. Alternatively, 1 pound (454 g) of type II product shall be packaged and heat sealed in a laminated bag, one layer of which is foil, in a manner to protect it from deterioration and damage during shipment from the supply source to the first consuming activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. The product shall be packed in accordance with level A, B or C, as specified (see 6.1).

5.2.1 Level A packing. The type II product, packaged as specified in 5.1.1, shall be packed in accordance with level A packing requirements of PPP-C-29.

5.2.2 Level B packing. The type I or II product, packaged as specified in 5.1.1 or 5.1.2, shall be packed in accordance with level B packing requirements of PPP-C-29. Alternatively, a wax impregnated corrugated box, constructed, closed and reinforced in accordance with style RSC, class I, type SWCFI, grade 275 of PPP-B-1163 may be used.

5.2.2.1 Laminated bags in a waterproof case liner. When specified (see 6.1), twenty-four laminated bags of material shall be inclosed in a waterproof case liner constructed and closed in accordance with MIL-L-10547 for subsistence items. The product in the case liner shall be packed in a snug-fitting fiberboard box with liner constructed, closed, and reinforced in accordance with style RSC-L, V3c, V3s, or V4s of PPP-B-636. The waterproof case liner shall be protected from stitches in the bottom flaps with a piece of paperboard placed between the stitches and the case liner. For transfer at sea operations the shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636 except that two bands shall be used, one lengthwise and one girthwise.

5.2.3 Level C packing. The type I or II product, in cans, packaged as specified in 5.1.3, shall be packed in accordance with level C packing requirements of PPP-C-29. Type II product in laminated bags shall be packed in accordance with the Uniform Freight Classification or the National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Marking.

5.4.1 Labeling of cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use shall be included in the labeling:

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Directions for use - Type I Product

(1) Defrost topping to a temperature of 40°F to 45°F.

(2) Empty contents of can into chilled mixing bowl.

(3) Whip with wire beater on medium speed (2nd) until topping forms smooth stiff peaks, about 10 to 20 minutes.

Yield: 1-3/4 to 2 gal. whipped topping.

PERISHABLE - KEEP FROZEN

(0°F or below)

Directions for use - Type II Product
As a Whipped Topping

(Follow manufacturer's instructions)

As a Pastry Filling

Reduce the amount of water used for reconstitution by 15-20 percent or approximately 1/2 cup for each pound of dehydrated product.

Dehydrated topping should be at room temperature (60°F to 80°F). Add contents of can (1 pound) to 1 quart of cold (40°F to 45°F) reconstituted nonfat dry milk 1/ in a chilled mixer bowl. Temper contents to 40°F to 45°F. Blend for three minutes at low speed.

Scrape down bowl and beater. Whip mixture at high speed until stiff peaks are formed (5 to 10 minutes).

Cover and refrigerate until ready for use.

Yield: 1 gal. or 50 portions of 1/3 cup.

1/ Use 3-1/4 ounces nonfat dry milk and 3-3/4 cups of cold water 40°F to 45°F.

5.4.2 Laminated bags. Laminated bags shall be labeled in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Labeling shall include the manufacturer's directions for use.

5.4.3 Marking of shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary markings shall be stencilled or printed on top of the boxes, for the type product indicated, in letters not less than 1-1/2 inches (38.0 mm) high:

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For Type I Product: KEEP FROZEN
(0°F (-18°C) OR BELOW)

For Type II Product: STORE IN COOL, DRY AREA

5.4.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of product required (see 1.2.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Type and class of unit load when unit loading is specified (see 5.3).
(e) Laminated bags in a waterproof case liner when packaging specified in 5.2.2.1 is required.

6.2 Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

6.3 Intended use of product and technical information. The products covered in this specification are a satisfactory substitute for whipped cream as a topping for desserts and baked items. They can be combined with fruits, nuts, flavoring and confectionary products to make a variety of desserts. Compared to dairy whipping cream, there is less chance of churning due to over-beating and less serum separation in the whipped product when exposed to warm temperatures. The dehydrated product has good stability and need not be held under refrigerated storage.

6.4 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.14D or/DSAR 4145.7, as applicable.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for

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any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - G1
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0358

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-43856A
TOPPING, DESSERT AND BAKERY PRODUCTS, FROZEN OR DEHYDRATED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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